

Final Project Report

ComicCraft – AI Comic Story Creator

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Team ID: SWTID-2026-2688

Team Size: 5

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1. Introduction

ComicCraft – AI Comic Story Creator is a proposed web application that helps users transform a short story prompt into a multi-panel comic draft with generated illustrations and PDF export. It targets creators who want to communicate visual stories without requiring advanced drawing or layout skills.

2. Ideation Phase

The identified user problem is the time and expertise required for scripting, illustration, and page layout. Brainstorming prioritized prompt-to-panel outline generation, AI-assisted illustration, a responsive FastAPI/Jinja2 interface, and PDF output. The empathy map highlights learning curve and time as key pains, with accessible generation and creative freedom as desired gains.

3. Requirement Analysis

Core functional requirements are user input, prompt processing, content and image generation, and PDF export. Usability, secure credential handling, reliability, and low latency are key non-functional requirements. The proposed stack includes Python, FastAPI, Uvicorn, Jinja2, Gemini models, Diffusers/Stable Diffusion, dotenv, and FPDF.

4. Project Design

The architecture routes browser input through FastAPI to the configured generation services. Generated scripts and images pass through a layout builder before PDF compilation. Input validation, server-side secret handling, timeouts, and graceful failure paths are design considerations.

5. Project Planning & Scheduling

Three planned sprints cover ideation/design (5 points), backend and AI integration (8 points), and frontend/PDF export (8 points). The schedule assigns named team members as specified in the planning template; implementation progress should be recorded against backlog items.

6. Functional & Performance Testing

The supplied UAT summary reports 61 of 61 cases passed across PDF generation (7), interface (51), and security/token handling (3). Functional scenarios cover validation, generation, API connectivity, and response time. Actual timings and detailed execution evidence were not included and should be appended before claiming verified performance.

7. Output Screenshots & Conclusion

Insert actual application screenshots, generated comic samples, and PDF output here. The concept can reduce barriers to producing a comic draft, but generated imagery may vary and editing or character consistency may require additional controls.

Advantages, limitations & future scope

- Advantages: guided workflow, reduced manual layout effort, rapid draft creation, downloadable output.
- Limitations: dependence on external AI services, variable generation quality and latency, possible provider costs, and unverified character consistency.
- Future scope: panel editing, regeneration controls, style presets, stronger character references, accessibility improvements, background job/progress handling, and broader user validation.

Repository and evidence

GitHub repository: <https://github.com/Annamalai13/ComicCraft---AI-Comic-Story-Creator-using-Gemini-Models.git>

Video Link : <https://drive.google.com/drive/folders/1hY-CfrwPSgOxbgWkvRJ304Hicxi-E7fB?usp=sharing>

Add deployment instructions, version/commit, screenshots, test logs, and authorization for any included user or provider data.